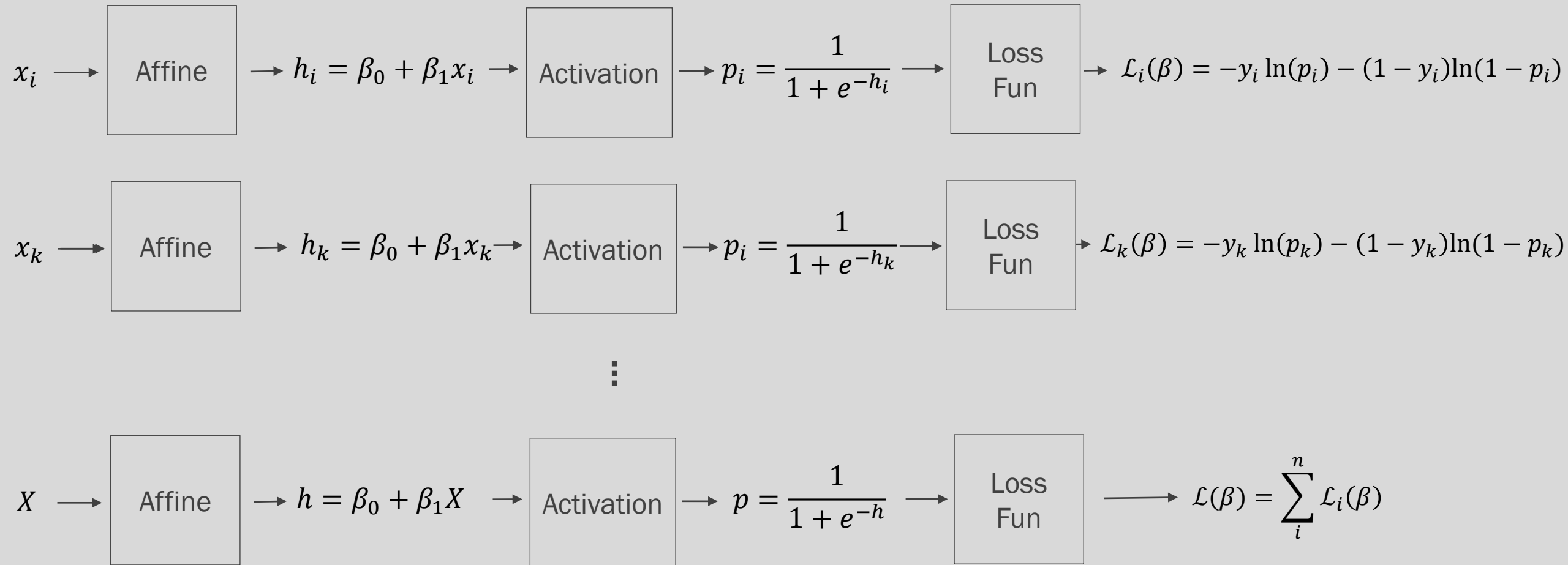




Lecture Outline

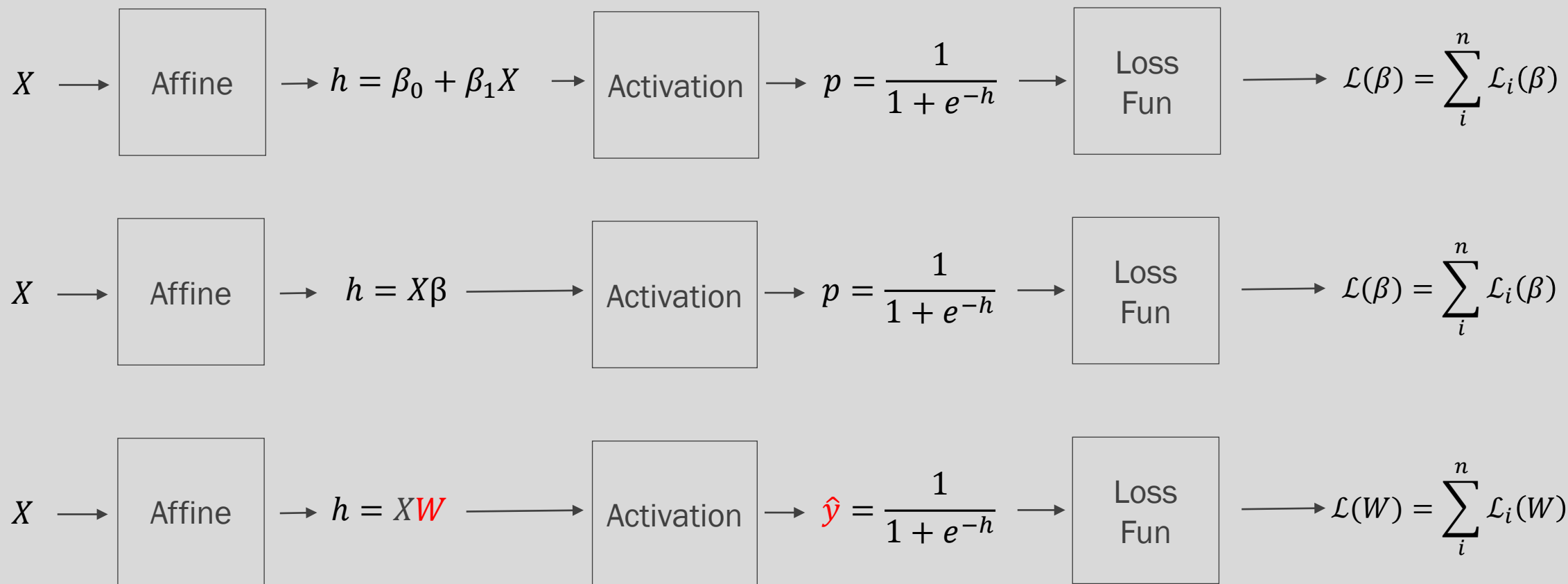
- Introduction to Artificial Neural Networks (ANNs)
- Review of Classification and Logistic Regression
- Single Neuron Network ('Perceptron')
- Multilayer Perceptron (MLP)

Logistic Regression Revisited



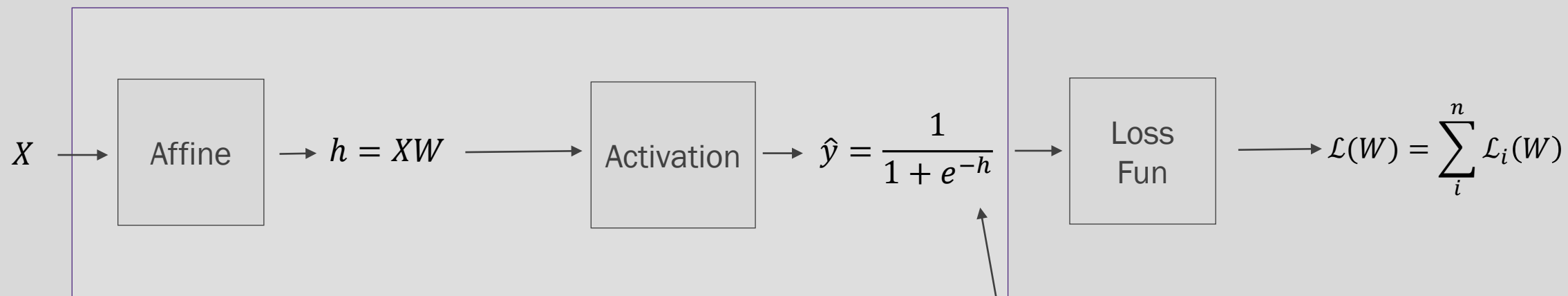


Our First ANN

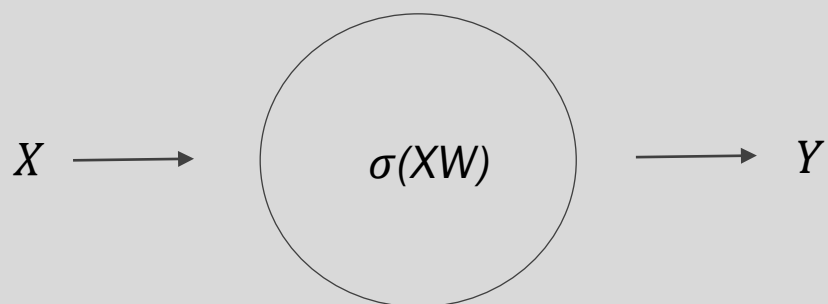




Our First ANN



“Sigmoid activation” σ



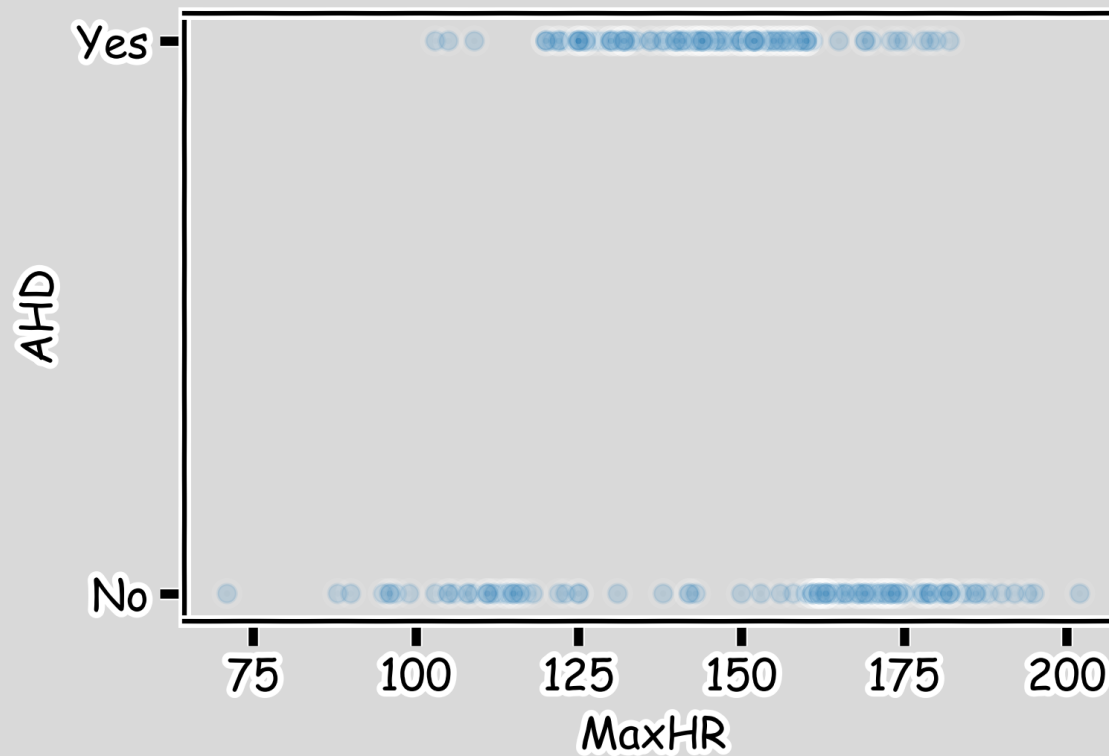
Logistic Regression can be thought of as a
Single Neuron Network!



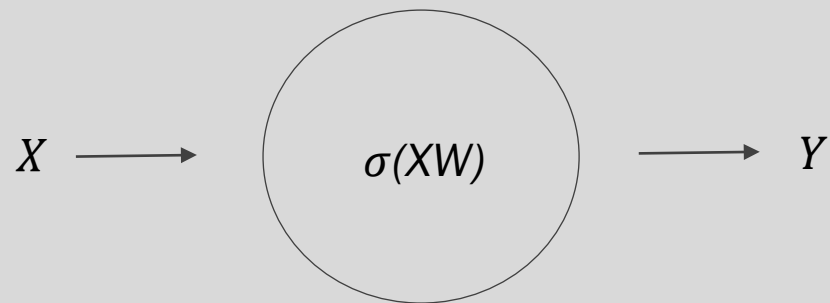
Lecture Outline

- Introduction to Artificial Neural Networks (ANNs)
- Review of Classification and Logistic Regression
- Single Neuron Network ('Perceptron')
- **Multilayer Perceptron (MLP)**

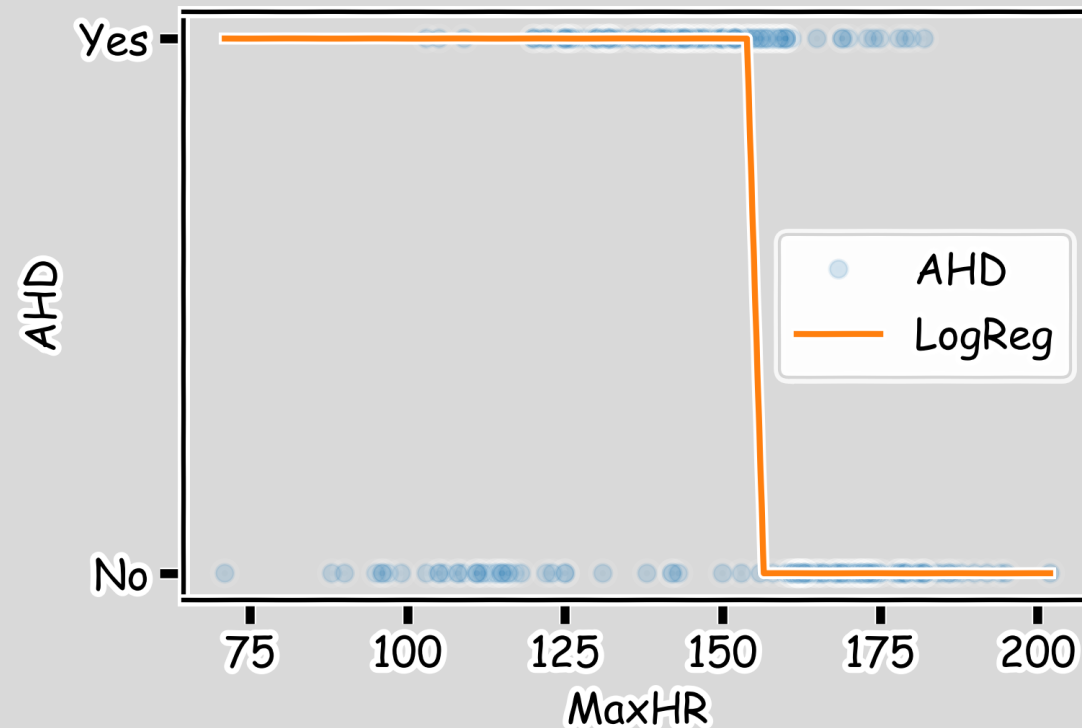
Example Using Heart Data



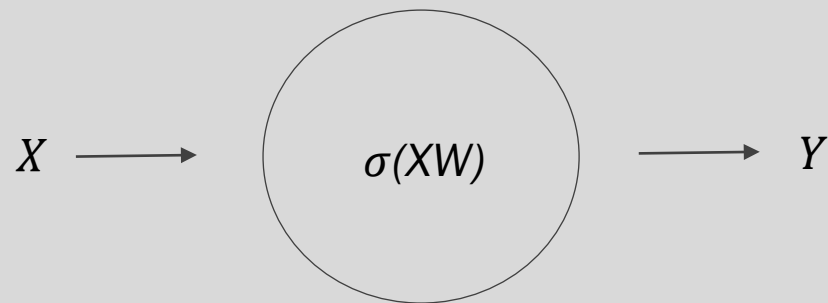
Example Using Heart Data



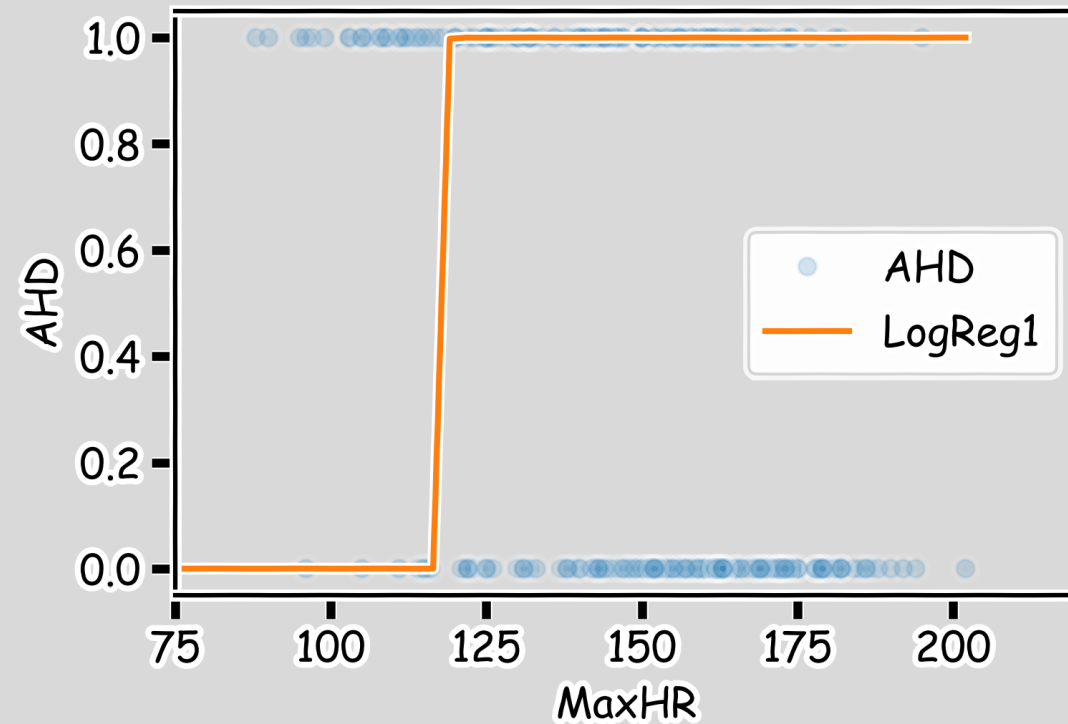
Choose W such that



Example Using Heart Data

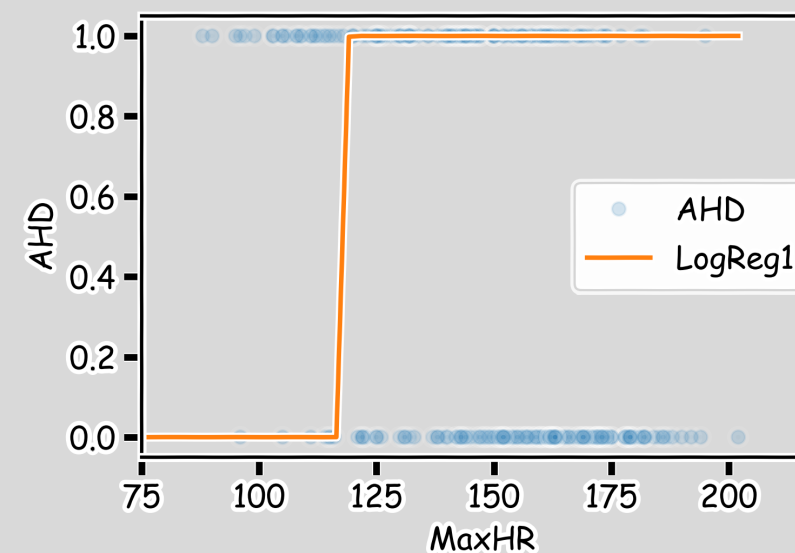
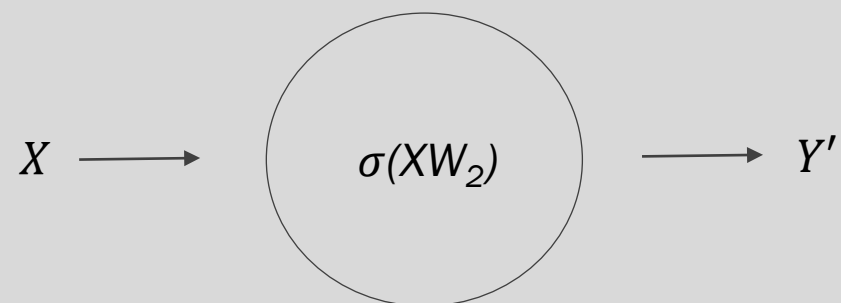
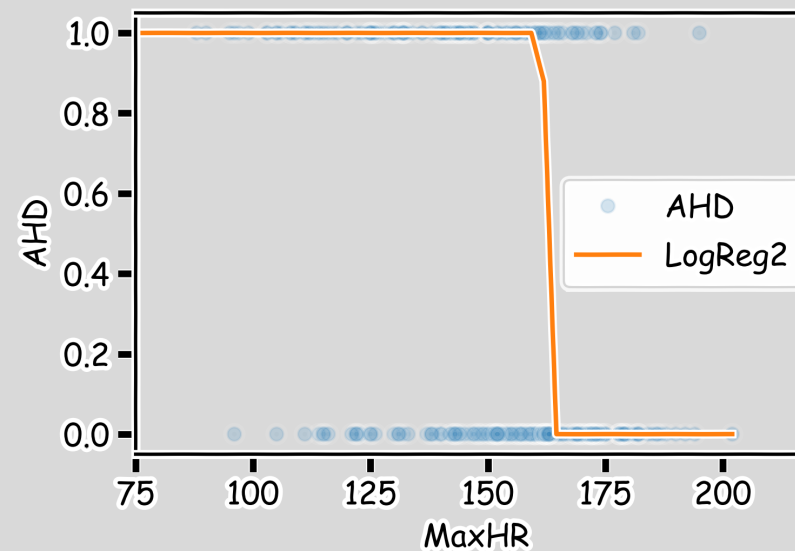
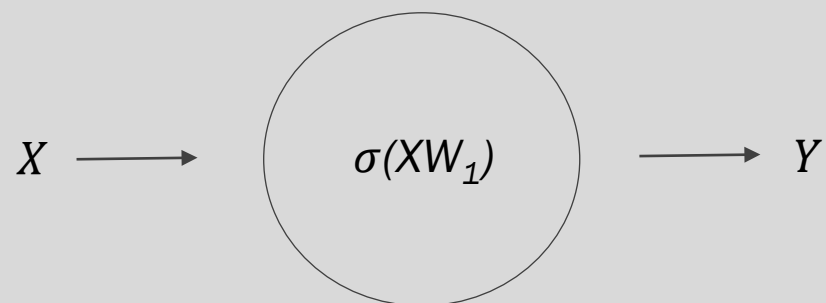


Choose W such that



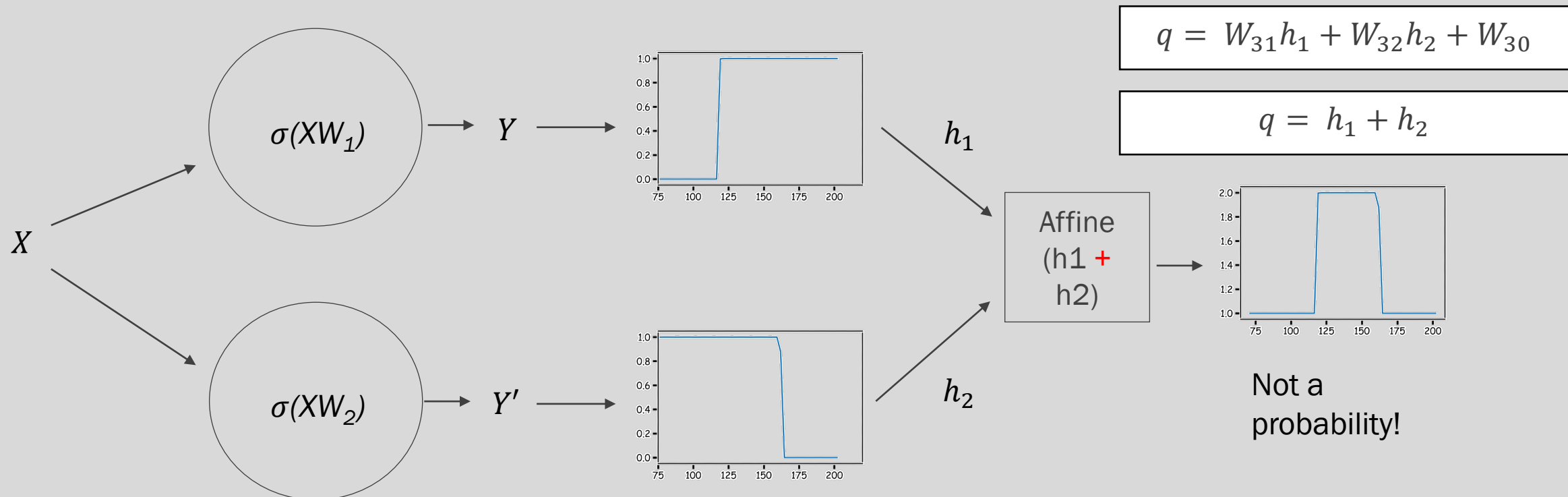


Example Using Heart Data



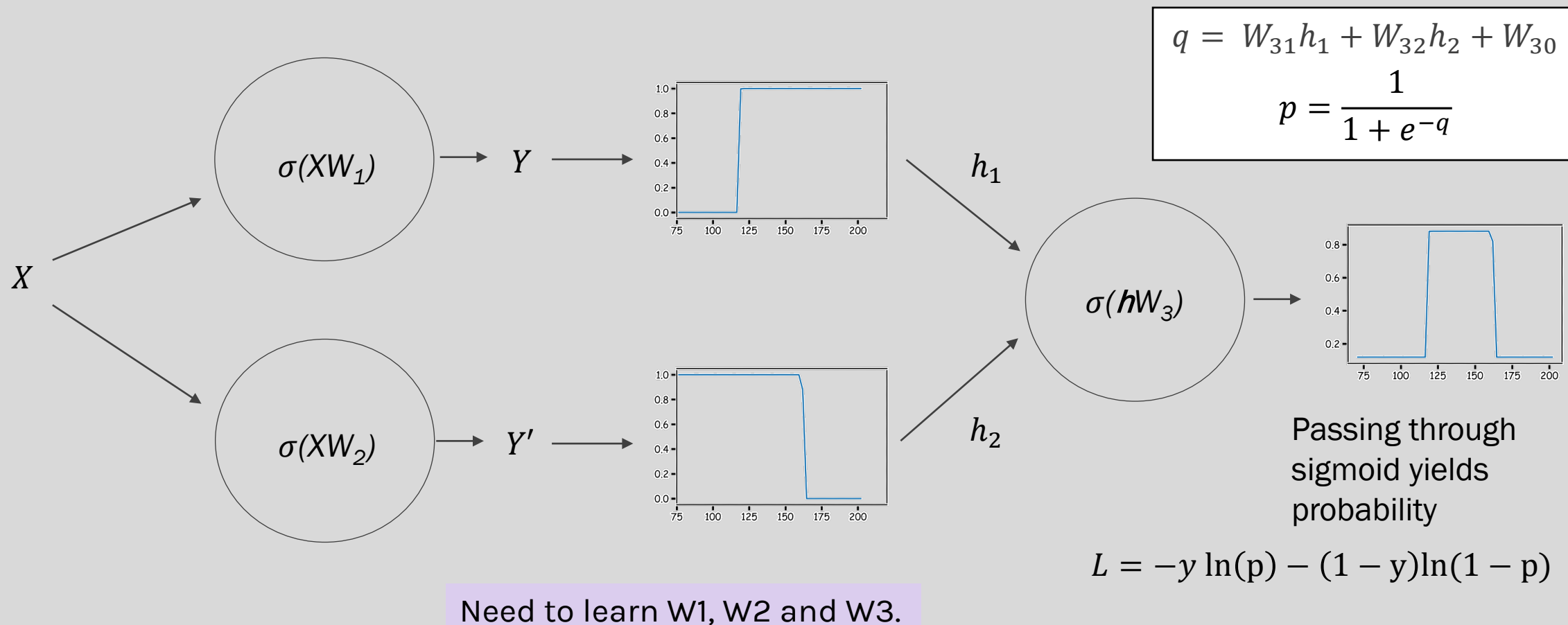


Combining Neurons

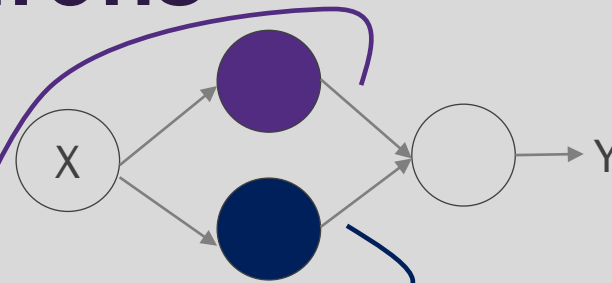




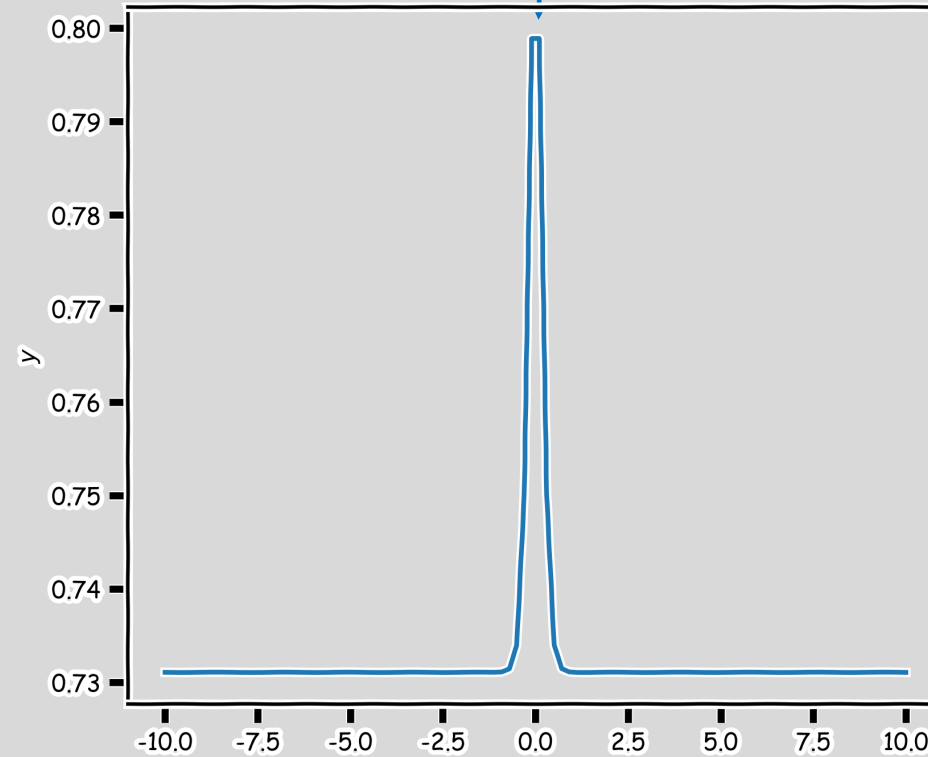
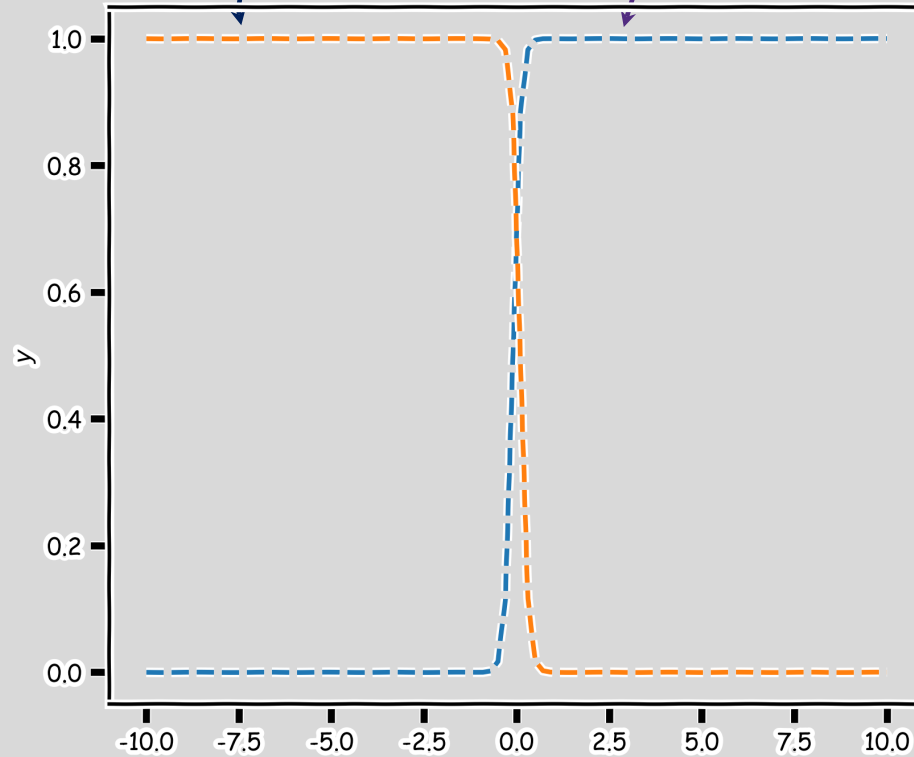
Combining Neurons



Combining Neurons

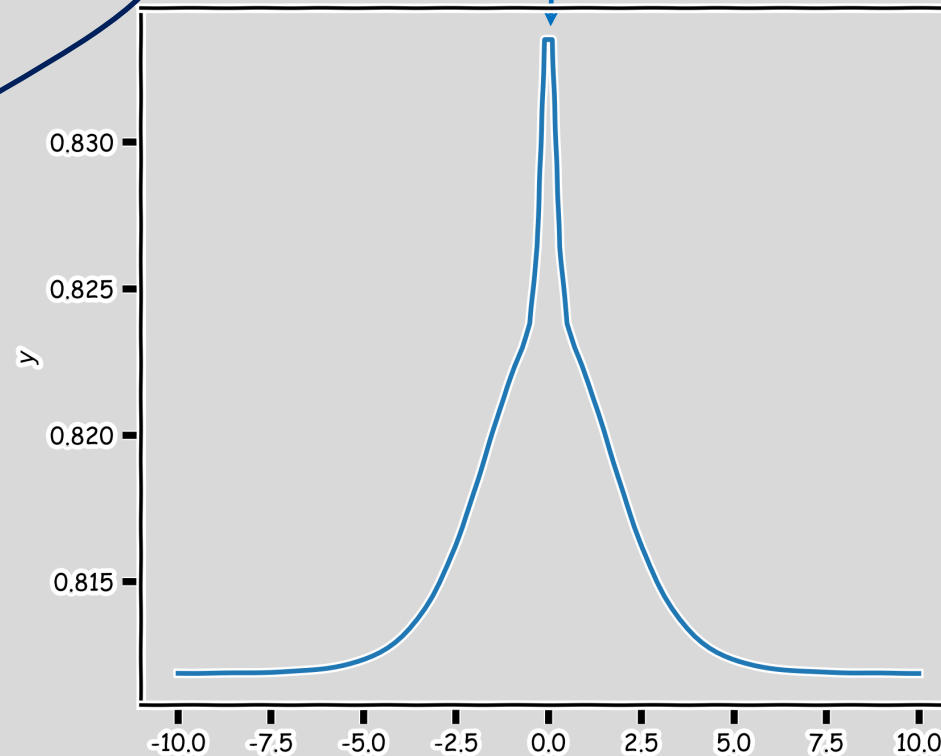
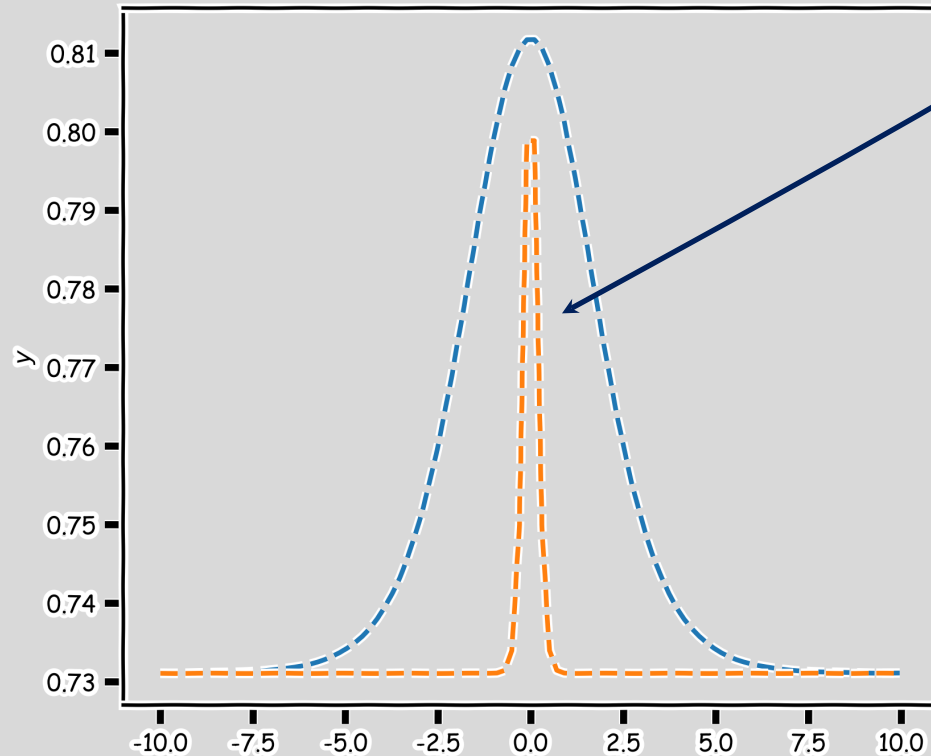
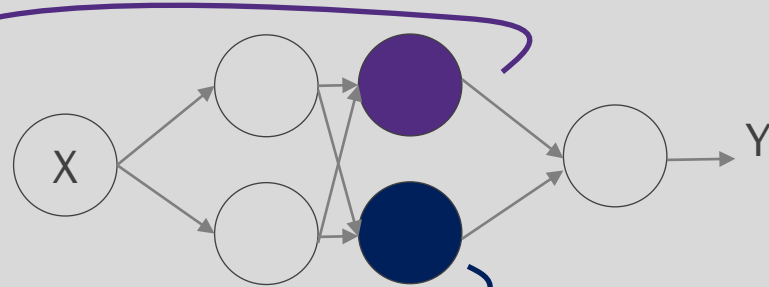


Different weights change the shape and position

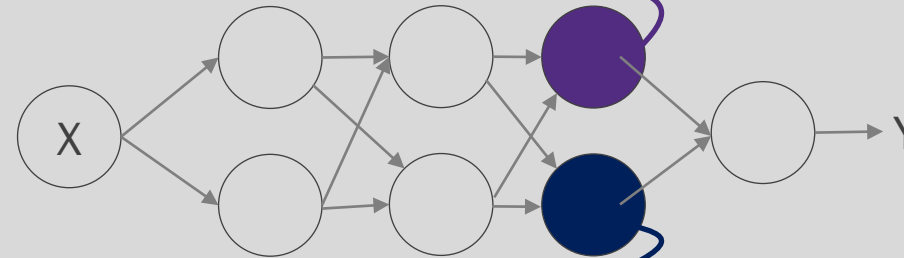


Combining Neurons

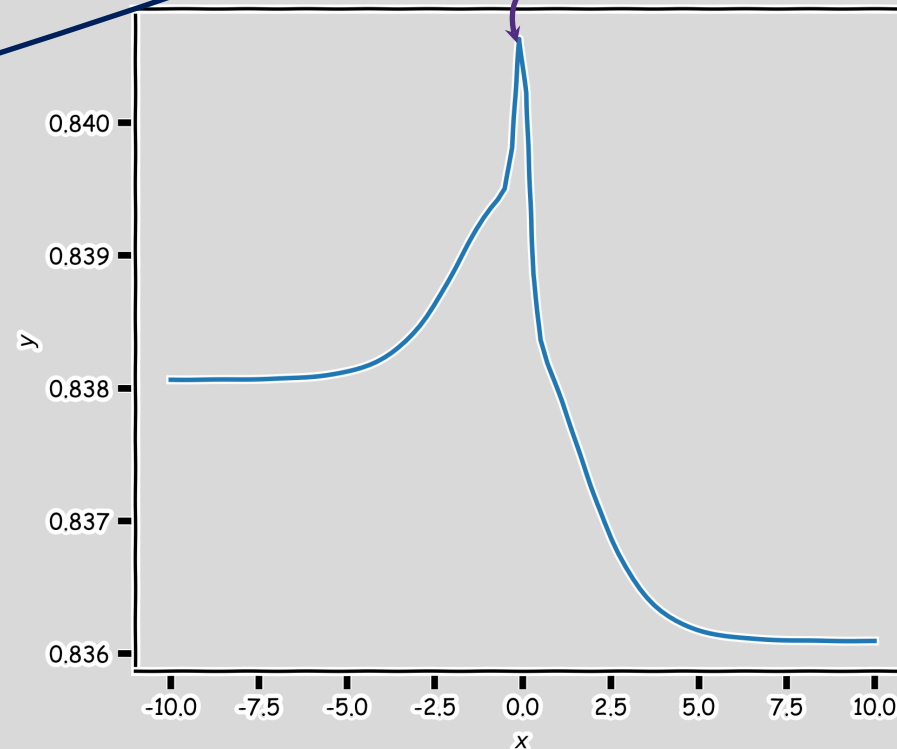
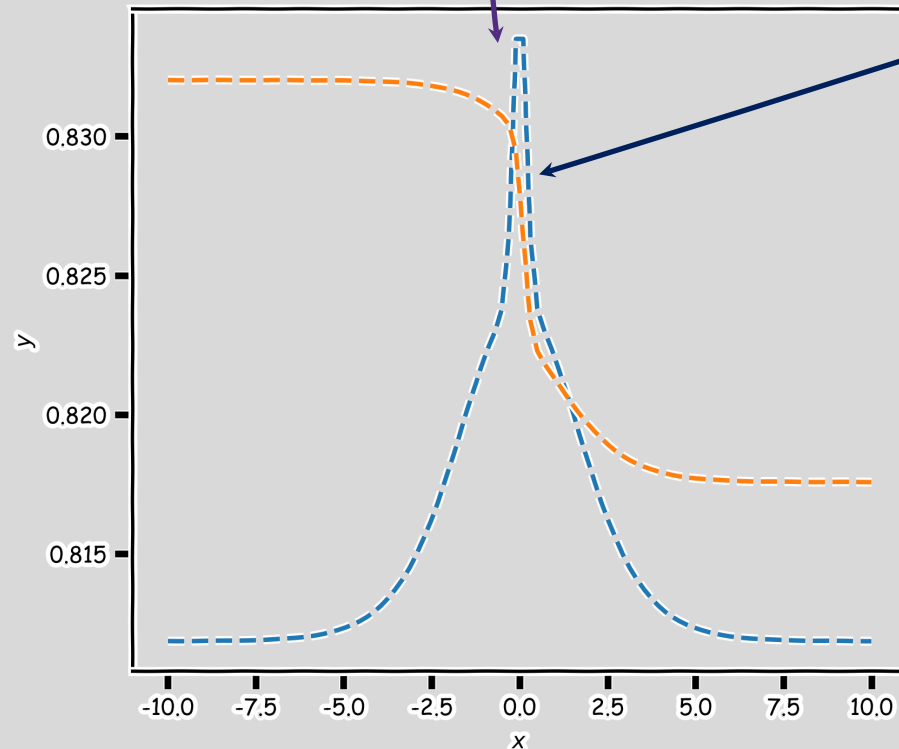
Neural networks can model *any* reasonable function!



Combining Neurons



Adding layers allows us to model increasingly complex functions





Summary

So far:

- A single neuron can be a logistic regression unit. We will soon see other choices.
- A neural network is a combination of logistic regression (or other types) units.
- A neural network can approximate non-linear functions.

Coming Soon:

- What kind of activations, how many neurons, how many layers, output unit and loss function?
- How do we estimate the weights and biases?
- How to regularize Neural Networks.



Questions?